Richard B. Anderson

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EDUCATION

The University of Texas at El Paso, El Paso, TX *May 2026*
Bachelor of Science in Physics, Magna Cum Laude
Bachelor of Science in Mathematics, Magna Cum Laude

- GPA: 3.81 / 4.00

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Fortran, Rust, Bash, Mathematica
Scientific Computing and Data: NumPy, SciPy, Pandas, Matplotlib, HDF5, TensorFlow, OpenCV
Modeling and Simulation: OpenMC, LIGGGHTS, Geant4, REBOUND, Simcenter STAR-CCM+, ParaView, MATLAB, Simulink
High-Performance Computing: Linux, Unix, SLURM, MPI, OpenMP, CUDA, parallel computing, computer hardware troubleshooting and maintenance
Engineering and Development Tools: Siemens NX, Teamcenter, ArduPilot, UAV hardware integration, CAD, Git, CMake, Make, GDB, Valgrind, Docker

TEACHING EXPERIENCE

Undergraduate Teaching Assistant, Electromagnetism Laboratory *El Paso, TX*
The University of Texas at El Paso, Department of Physics *Aug 2025 – Present*

- Guide students through electromagnetism lab setup, data collection, and post-lab data analysis.
- Troubleshoot experimental and conceptual problems while connecting measurements to course concepts.

Academic Tutor *El Paso, TX*
El Paso Community College, Student Success *Jul 2022 – Aug 2023*

- Tutored college mathematics from remedial algebra through differential equations in one-on-one and online settings.
- Adapted explanations to varied preparation levels, laying out step-by-step reasoning and problem setup.
- Maintained session records and coordinated with staff during high-demand periods.

TECHNICAL EXPERIENCE

Undergraduate Research Assistant *El Paso, TX*
UTEP Astrophysics Theory Group | Advisor: Dr. Kedron Silsbee *Aug 2022 to May 2026*

- Developed semi-analytic and numerical models of protostellar collapse and planetesimal capture using Python, C, and the REBOUND N-body simulation framework.
- Implemented custom numerical integration, interpolation, and ordinary-differential-equation solvers for analytically intractable physical models.
- Designed SLURM-based workflows for distributed simulation execution, automated data collection, and analysis of high-volume outputs on UTEP's high-performance computing cluster.
- Diagnosed numerical instability, verified simulation results, and built post-processing tools for orbital evolution, capture outcomes, and statistical analysis.
- Prepared a research manuscript and presented results at the APS Texas Section Fall Meeting in 2025.

SULI Intern, Nonproliferation and National Security Department *Upton, NY*
Brookhaven National Laboratory *Jun 2025 to Aug 2025*

- Developed an end-to-end computational workflow integrating LIGGGHTS discrete-element models, OpenMC radiation-transport simulations, CAD, Python data processing, and machine learning.
- Generated stochastic pebble-bed reactor geometries and incorporated them into Monte Carlo simulations to calculate neutron and gamma-ray spectra across fuel-burnup conditions.
- Built Python pipelines for simulation execution, data reduction, feature extraction, visualization, and preparation of machine-learning inputs.
- Trained TensorFlow models to estimate fuel burnup and reactor operating time from simulated gamma-ray spectra for nuclear-safeguards research.

Undergraduate Research Intern *Arlington, TX*
UT Arlington | Neutrino and Rare Event Searches *May 2024 to Dec 2024*

- Developed Python waveform-analysis pipelines for NEXT detector data to extract decay constants, peak widths, drift-time dependence, and pulse-shape observables.
- Applied SciPy signal-processing methods to denoise experimental waveforms, identify signal features, and compare reconstructed observables across operating conditions.
- Generated and analyzed alpha-particle events in Geant4/NEXUS to evaluate detector response, ionization transport, spatial distributions, and background signatures.
- Used the nuDoBE framework and Kolmogorov-Smirnov statistics to compare energy and angular distributions predicted by neutrinoless double-beta-decay models.

PUBLICATIONS AND PRESENTATIONS

Richard B. Anderson and Kedron Silsbee: Capture of Interstellar Planetesimals. Manuscript in preparation
Richard B. Anderson and Kedron Silsbee: Capture of Freely Floating Planetesimals Into Protoplanary Disks. APS Texas Section Fall Meeting, 2025

HONORS AND AWARDS

Outstanding Senior in Physics: UTEP Department of Physics, 2026
NSF Graduate Research Fellowship Program Honorable Mention: National Science Foundation, 2026
Terry Foundation Scholarship: Terry Foundation, 2023 to 2027
NuPEERS Nuclear Physics Summer School: NuPEERS, 2023